Homework Template

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November 3, 2025

Problem 1

What time does a 12-hour clock read:

1. 80 hours after 11:00?
2. 40 hours before 12:00?
3. 100 hours after 6:00?

Solution. Reduce offsets modulo 12 hours.

1. $80 \equiv 8 \pmod{12}$, so $11{:}00 + 8$ hours $= 19{:}00 \equiv \boxed{7{:}00}$.
2. $40 \equiv 4 \pmod{12}$, so 40 hours before is 4 hours before: $12{:}00 - 4 = \boxed{8{:}00}$.
3. $100 \equiv 4 \pmod{12}$, so $6{:}00 + 4 = \boxed{10{:}00}$.

Problem 2

Prove: if a is a positive integer, then $4 \nmid (a^2 + 2)$.

Solution. Consider a modulo 4.

If a is even, write $a = 2k$. Then $a^2 + 2 = 4k^2 + 2 \equiv 2 \pmod{4}$, not divisible by 4.

If a is odd, write $a = 2k + 1$. Then $a^2 + 2 = 4k^2 + 4k + 3 \equiv 3 \pmod{4}$, not divisible by 4.

In all cases $a^2 + 2 \not\equiv 0 \pmod{4}$, so $4 \nmid (a^2 + 2)$.

Problem 3

Prove: if n is an odd positive integer, then $n^2 \equiv 1 \pmod{8}$.

Solution. Let $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k(k + 1) + 1.$$

One of k or $k + 1$ is even, so $k(k + 1) = 2m$ for some integer m. Hence

$$n^2 = 4 \cdot 2m + 1 = 8m + 1 \equiv 1 \pmod{8}.$$

Problem 4

Use the extended Euclidean algorithm to express the following gcds as linear combinations.

1. $\gcd(26, 91)$.
2. $\gcd(252, 356)$.
3. $\gcd(144, 89)$.
4. $\gcd(1001, 100001)$.

Solution.

1. $91 = 3 \cdot 26 + 13$, $26 = 2 \cdot 13$. Hence
$$13 = 91 - 3 \cdot 26.$$
2. $356 = 1 \cdot 252 + 104$, $252 = 2 \cdot 104 + 44$, $104 = 2 \cdot 44 + 16$, $44 = 2 \cdot 16 + 12$, $16 = 1 \cdot 12 + 4$. Back-substitute:
$$4 = 17 \cdot 356 - 24 \cdot 252.$$
3. Euclid: $144 = 1 \cdot 89 + 55$, $89 = 1 \cdot 55 + 34$, $55 = 1 \cdot 34 + 21$, $34 = 1 \cdot 21 + 13$, $21 = 1 \cdot 13 + 8$, $13 = 1 \cdot 8 + 5$, $8 = 1 \cdot 5 + 3$, $5 = 1 \cdot 3 + 2$, $3 = 1 \cdot 2 + 1$. Back-substitute to get
$$1 = 34 \cdot 144 - 55 \cdot 89.$$
4. $100001 = 99 \cdot 1001 + 902$, $1001 = 1 \cdot 902 + 99$, $902 = 9 \cdot 99 + 11$, $99 = 9 \cdot 11$. Back-substitute:
$$11 = 10 \cdot 100001 - 999 \cdot 1001.$$

Problem 5

Among any three consecutive odd integers $p, p + 2, p + 4$, exactly one is divisible by 3.

1. Find a triple of the form $p, p + 2, p + 4$ consisting entirely of primes.
2. Prove there is no other triple of three consecutive odd positive integers that are all prime.

Solution.

1. $\boxed{3, 5, 7}$.
2. For any odd p, the three numbers are congruent to one of $\{0, 2, 1\}$ modulo 3 in some order, so one is divisible by 3. If all three are prime, the divisible one must equal 3 itself; otherwise it is composite. Hence the only possibility is the triple containing 3, namely $3, 5, 7$.

Problem 6

Prove that the product of any three consecutive integers is divisible by 6.

Solution. Let the integers be $n, n + 1, n + 2$.

At least one of them is even, so the product is divisible by 2. Also, one of $n, n + 1, n + 2$ is divisible by 3 (pigeonhole principle modulo 3). Therefore the product is divisible by both 2 and 3, hence by $\operatorname{lcm}(2, 3) = 6$.

Problem 7

Solve each congruence.

1. $34x \equiv 77 \pmod{89}$.

   Since $\gcd(34, 89) = 1$, 34 has an inverse modulo 89. From

   $$1 = 13 \cdot 89 - 34 \cdot 34,$$

   the inverse of 34 modulo 89 is $-34 \equiv 55$. Hence

   $$x \equiv 55 \cdot 77 \equiv 4235 \equiv \boxed{52} \pmod{89}.$$

2. $144x \equiv 4 \pmod{233}$.

   Using the extended Euclidean algorithm,

   $$1 = 89 \cdot 144 - 55 \cdot 233,$$

   so the inverse of 144 modulo 233 is 89. Therefore

   $$x \equiv 89 \cdot 4 \equiv 356 \equiv \boxed{123} \pmod{233}.$$

3. $200x \equiv 13 \pmod{1001}$.

   Since $1001 \equiv 1 \pmod{200}$, we have $200 \cdot (-5) \equiv -1000 \equiv 1 \pmod{1001}$, so $200^{-1} \equiv -5 \equiv 996 \pmod{1001}$. Thus

   $$x \equiv 996 \cdot 13 \equiv 12{,}948 \equiv \boxed{936} \pmod{1001}.$$

Problem 8

Show that for every positive integer n, $42 \mid (n^7 - n)$.

Solution. Factor $42 = 2 \cdot 3 \cdot 7$. It suffices to show divisibility by 2, 3, and 7.

Modulo 2. n and n^7 have the same parity, so $n^7 - n$ is even.

Modulo 3. If $3 \mid n$ we are done. Otherwise, by Fermat's little theorem $n^2 \equiv 1 \pmod{3}$, hence $n^6 \equiv 1$ and

$$n^7 - n = n(n^6 - 1) \equiv n(1 - 1) \equiv 0 \pmod{3}.$$

Modulo 7. If $7 \mid n$ we are done. Otherwise, Fermat gives $n^6 \equiv 1 \pmod{7}$, so $n^7 \equiv n$ and therefore $n^7 - n \equiv 0 \pmod{7}$.

Thus $n^7 - n$ is divisible by 2, 3, and 7, hence by 42.

Problem 9

1. Compute 5^{2003} mod 7, mod11, and mod13.

   By Fermat:

   $$5^6 \equiv 1 \ (\text{mod}7) \Rightarrow 5^{2003} \equiv 5^5 \equiv 3 \ (\text{mod}7),$$
   $$5^{10} \equiv 1 \ (\text{mod}11) \Rightarrow 5^{2003} \equiv 5^3 \equiv 125 \equiv 4 \ (\text{mod}11),$$
   $$5^{12} \equiv 1 \ (\text{mod}13) \Rightarrow 5^{2003} \equiv 5^{11} \equiv 5^{-1} \equiv 8 \ (\text{mod}13).$$

2. Use (a) and the Chinese remainder theorem to find $5^{2003} \bmod 1001$.

   Solve
   $$x \equiv 3 \pmod{7}, \qquad x \equiv 4 \pmod{11}, \qquad x \equiv 8 \pmod{13}.$$
   Combine the first two: $x \equiv 59 \pmod{77}$. Now solve $59 + 77t \equiv 8 \pmod{13}$. Since $59 \equiv 7$ and $77 \equiv 12 \equiv -1 \pmod{13}$, we get $7 - t \equiv 8$, so $t \equiv -1 \equiv 12$. Thus $x = 59 + 77 \cdot 12 = 983$ is a solution modulo $1001 = 7 \cdot 11 \cdot 13$.

   Therefore,
   $$\boxed{5^{2003} \equiv 983 \pmod{1001}}.$$

Problem 10

Find the number of trailing zeros in 100!.

Solution. The number of zeros is the exponent of $10 = 2 \cdot 5$ in 100!, which is $\min\{v_2(100!), v_5(100!)\} = v_5(100!)$.

$$v_5(100!) = \left\lfloor \frac{100}{5} \right\rfloor + \left\lfloor \frac{100}{25} \right\rfloor + \left\lfloor \frac{100}{125} \right\rfloor = 20 + 4 + 0 = 24.$$

So 100! ends with $\boxed{24}$ zeros.

Problem 11

Use the generalized Chinese Remainder Theorem.

(a)

Solve
$$x \equiv 5 \pmod{6}, \qquad x \equiv 3 \pmod{10}, \qquad x \equiv 8 \pmod{15}.$$
Consistency checks:
$$\begin{aligned} \gcd(6,10) = 2: &\quad 5 \equiv 3 \equiv 1 \pmod{2} \quad \checkmark \\ \gcd(6,15) = 3: &\quad 5 \equiv 8 \equiv 2 \pmod{3} \quad \checkmark \\ \gcd(10,15) = 5: &\quad 3 \equiv 8 \equiv 3 \pmod{5} \quad \checkmark \end{aligned}$$
So a solution exists and is unique modulo $\operatorname{lcm}(6,10,15) = 30$.

From $x \equiv 5 \pmod{6}$ the candidates modulo 30 are $5, 11, 17, 23, 29$. Among these, those $\equiv 3 \pmod{10}$ are 23. Check $23 \equiv 8 \pmod{15}$, true.

$$\boxed{x \equiv 23 \pmod{30}}.$$

(b)

Solve
$$x \equiv 7 \pmod{9}, \qquad x \equiv 4 \pmod{12}, \qquad x \equiv 16 \pmod{21}.$$
Consistency checks (all use gcd = 3): $7 \equiv 4 \equiv 16 \equiv 1 \pmod{3}$, so consistent. Uniqueness is modulo $\operatorname{lcm}(9,12,21) = 252$.

First combine mod 9 and 12. Write $x = 7 + 9a$. Then $7 + 9a \equiv 4 \pmod{12} \iff 9a \equiv 9 \pmod{12}$, which reduces to $3a \equiv 3 \pmod{4} \iff a \equiv 1 \pmod{4}$. Thus $x \equiv 16 \pmod{36}$.

Now impose $x \equiv 16 \pmod{21}$ as well: $x = 16 + 36k \equiv 16 + 15k \pmod{21}$, so $15k \equiv 0 \pmod{21}$, which gives $k \equiv 0 \pmod{7}$. Hence $x = 16 + 252t$.

$$\boxed{x \equiv 16 \pmod{252}}.$$